Sizing a PV and BESS plant exposed to day-ahead spot marginal prices – A case study from Colombia

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Abstract

This document shows the results of a sizing model for a utility-scale PV plant located in Monteria, Cordoba, Colombia. The method determines the optimal size of the PV plant in MWdc (number of PV modules) and MW (size of the PV inverter) considering a fixed. The model also determines the size of the Battery Energy Storage System. The project is funded in the context of wholesale spot electricity markets where PV & BESS system's energy curtailed can be used to arbitrage. The method aims to get the PV plant dispatch (including curtailment) and the BESS scheduling such that maximize the Net Present Value of the project using Ember's economic figures [1]. The method takes advantage of BESS trading Colombian spot prices (2024) provided by XM in order to leverage the project. Results are obtained for two maximum NPV conditions in Monteria considering a point of interconnection of 100 MW: 1) determining the PV's MWdc by fixing a 80 MWh BESS system with C-rate of 0.25 (4 hours), and 2) determining PV's MWdc and BESS specification. For the first case, results show that to maximize the project NPV (445 million US$) it is necessary to install a 312 MW PV power plant with good financial KPI's: IRR of 27.6 %, a payback of 4.4 years, and a LCOE of 75.0 US$/MWh. For the second case, results show that to maximize the project NPV (690 million US$) it is necessary to install a 732 MW PV power plant with the following KPI's: IRR of 14.7 %, a payback of 9 years, and a LCOE of 123 US$/MWh.

Keywords: BESS, PV system, Energy Transition, Energy market, Power Dispatch, NPV, LCOE.

1. Methodology for BESS and PV sizing

The proposed method aims to determine PV & BESS component specifications such that they maximize the net present value of the project at a given capital recovery factor from the PV promoter perspective.

$$\max \text{NPV} = \frac{\text{PV \& BESS Project Cash Flow}}{CRFe} - CAPEX \tag{1}$$

The NPV metric incorporates capital cost of existing BESS assets and the size of the PV plant to find.

Variables and Dataset Definitions

The dataset containing all known data is listed as follows:

$CRF = \frac{i(i+1)^n}{(i+1)^n-1}$ is the capital recovery factor for a given project, where i is the discount rate in %, and n is the project lifespan in years.

e is the escalation factor in %.

$i_r = \frac{i-e}{1+e}$ is the modified discount rate in %.

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$CRF_e = \frac{i_r(i_r+1)^n}{(i_r+1)^n-1}(1+e)$ is the modified capital recovery factor for a given project including the escalation factor, where i_r is the modified discount rate in %.

$T = 8760$ hours is the time-project resolution for the power dispatch (one year).

C_{pv} is the PV system overnight cost in US$/kWp (includes PV modules, load-following inverters, balance of system, and installation).

C is the battery storage capacity overnight cost in US$/kWh.

C_{gfl} is the PV grid-following inverter overnight cost in US$/kWp.

C_{gfm} is the BESS grid-forming inverter overnight cost in US$/kWp.

$O\&M_{pv}$ is the PV system operation and maintenance cost in US$/kWp.

$O\&M_{bess}$ is the BESS operation and maintenance cost in US$/kWh.

$Services$ are additional incomes from services provided by the project, e.g., frequency and volt/var support, in US$/kWh.

S_c corresponds to additional costs of the project, such as balance of plant, balance of system and payments to obtain grid connection certification or permits from a national authority and final disposition of equipment at the end of the project period. In general, these costs range from 10% to 33% of the project CAPEX.

$\boldsymbol{\lambda} = [\lambda_1, \ldots, \lambda_t, \ldots, \lambda_T]$ is the set of hourly spot prices at the marketplace in US$/kWh. These prices can be forecasted or taken from historical databases (e.g., e year 2024).

η_c is the efficiency of the BESS inverter during charging in %.

η_d is the efficiency of the BESS inverter during discharging in %.

$P^u_{pv,t}$ is the average (2005-2023) hourly PV system output (DC power injection) in per unit for a given location. Hourly solar generation data is sourced from the European Commission's JRC PVGIS tool. Per unit data assumes fixed-tilt PV systems (crystalline silicon technology) with optimized slope and azimuth per location and a PV system loss of 19%. The ERAS5 solar irradiance database (2005-2023) is used for this purpose.

P^{max}_F is the maximum allowable power transfer at the point of interconnection (frontier) in kW.

$[\beta_{min}, \beta_{max}]$ are the battery operation limits in %. For a depth of discharge (DoD) of 100%, $\beta_{min} = 0.0$ and $\beta_{max} = 1.0$.

$[C^{min}_R, C^{max}_R]$ are the BESS inverter C-rate limits in 1/h. The C-rate indicates the time required to charge the battery. For instance, a C-rate of 2.0 implies the inverter can charge the battery in 30 minutes, while a C-rate of 0.1 implies a 10-hour charge time.

Proposed optimization model

The optimization problem is stated as finding the optimal cash-flow (optimal energy dispatch): $\mathbb{P}_s$, $\mathbb{P}_c$, $\mathbb{P}_d$, $\mathbb{P}^{max}_{pv}$, $\mathbb{P}_{pv}$, W_s, W_b, W_{pv}, W^{clip}_{pv}, $\mathbb{SOC}$, SOC_0, $\mathbf{w}$) and CAPEX (equipment specification: P^{inst}_{pv}, C_{bess}, $P^{inverter}_{bess}$, $P^{inverter}_{pv}$), such that the net present value of the project is maximized.

$$\max_{\substack{P^{inst}_{pv}, C_{bess}, P^{inverter}_{bess}, P^{inverter}_{pv}, \mathbb{P}_b \\ \mathbb{P}_s, \mathbb{P}_k, \mathbb{P}_c, \mathbb{P}_d, \mathbb{P}_{pv}, \mathbb{P}^{max}_{pv}, \mathbb{SOC}, SOC_0 \\ W_s, W_b, W_{pv}, W^{clip}_{pv} \\ \mathbf{w}}} \mathrm{NPV} = \frac{E_s - OPEX}{CRF_e} - CAPEX \tag{2}$$

subject to:

$$OPEX = O\&M_{pv} P_{pv}^{inst} + O\&M_{bess} C_{bess} \tag{3}$$

$$CAPEX = (C_{pv} P_{pv}^{inst} + C_{bess} C_{bess} + C_{gfm} P_{bess}^{inverter} + C_{gfl} P_{pv}^{inverter}) S_C \tag{4}$$

$$E_s = \sum_{t=1}^{T} \lambda_t P_{s,t} \tag{5}$$

$$P_{d,t} + P_{b,t} + P_{pv,t} = P_{c,t} + P_{s,t}, \forall t = 1, ..., T \tag{6}$$

$$P_{pv,t}^{max} = P_{pv,t}^{u} P_{pv}^{inst}, \forall t = 1, ..., T \tag{7}$$

$$SOC_t = SOC_{t-1} + P_{c,t}\eta_c - \frac{P_{d,t}}{\eta_d}, \forall t > 1, \quad SOC_1 = SOC_0 + P_{c,1}\eta_c - \frac{P_{d,1}}{\eta_d} \tag{8}$$

$$P_{pv}^{inverter} = P_F^{max} + P_{bess}^{inverter} \tag{9}$$

$$P_{pv,t} \leq P_{pv,t}^{max}, \forall t = 1, ..., T \tag{10}$$

$$P_{c,t} \leq P_{bess}^{inverter} w_t, \forall t = 1, ..., T \text{ [Mixed-Integer (Binary) Non Linear constraint]} \tag{11}$$

$$P_{d,t} \leq P_{bess}^{inverter}(1 - w_t), \forall t = 1, ..., T \text{ [Mixed-Integer (Binary) Non Linear constraint]} \tag{12}$$

$$P_{s,t} \leq P_F^{max}, \forall t = 1, ..., T \tag{13}$$

$$\beta_{min} C_{bess} \leq SOC_t \leq \beta_{max} C_{bess}, \forall t = 0, 1, ..., T \tag{14}$$

$$C_R^{min} C_{bess} \leq P_{bess}^{inverter} \leq C_R^{max} C_{bess} \tag{15}$$

$$W_s = \sum_{t=1}^{T} P_{s,t}, \quad W_b = \sum_{t=1}^{T} P_{b,t}, \quad W_{pv} = \sum_{t=1}^{T} P_{pv}^{t}, \quad W_{pv}^{clip} = \sum_{t=1}^{T} P_{pv,t}^{max} - W_{pv} \tag{16}$$

The model is formulated as an equivalent 8760-hour dispatch to determine the Net Present Value (NPV) over n years at a given discount rate i_r. To incorporate degradation effects and mid-life replacements for PV and battery components, the model can be expanded to a multi-year stream of $n \times 8760$ hours. For clarity, however, the following results are presented using discounted energy and costs based on a standard 8760-hour timeframe. The optimization model consists of $26 + P + 9T$ unknowns, $6 + 3T$ equality constraints, and $2 + 8T + P$ inequality constraints. While the number of variables exceeds the number of equality constraints providing degrees of freedom the inclusion of numerous inequality constraints significantly restricts the feasible region.

The unknown variables are grouped into PV & BESS equipment specifications, economic variables, and power/energy dispatch results.

Equipment Specification

P_{pv}^{inst} is the PV system capacity in kWdc.

C_{bess} is the battery capacity in kWh.

$P_{bess}^{inverter}$ is the BESS inverter capacity in kW.

$P_{pv}^{inverter}$ is the PV inverter capacity in kWp.

For instance, the number of modules (N_{pv}) for a 500 W crystalline silicon module, assuming a maximum irradiance of 1000 W/m^2, is:

$$N_{pv} = \frac{P_{pv}^{inst} \cdot 1000}{1000 \cdot A \cdot \eta} \tag{17}$$

where $A = 2.4\,\text{m}^2$ (single PV module area) and $\eta = 20.94\%$ is the single module efficiency.

Economic Variables to be optimized:

CashFlow $= E_s - OPEX$ is the cash flow of the PV & BESS project

Benefit $= E_s +$ Services $- OPEX$ represents the annual project benefit in US$/year.

OPEX : Operational expenditure for energy purchased from the market with the PV & BESS system installed (US$/year).

Services: Additional income obtained by providing system services (US$/year).

CAPEX: Capital expenditure associated with the investment in PV & BESS equipment (US$).

E_s: Income from excess energy sold to the market (US$/year).

Optimal Power and Energy Dispatch Variables

$\mathbb{P}_{pv} = [P_{pv,1}, ..., P_{pv,t}, ..., P_{pv,T}]$ where $P_{pv,t}$ is the power injected by the PV inverter (AC) at the hour t in kW.

$\mathbb{P}_{pv}^{max} = [P_{pv,1}^{max}, ..., P_{pv,t}^{max}, ..., P_{pv,T}^{max}]$ where $P_{pv,t}^{max}$ is the power generated by the PV system (DC) at the hour t in kW.

$\mathbb{P}_s = [P_{s,1}, ..., P_{s,t}, ..., P_{s,T}]$ where $P_{s,t}$ is the power sold to the grid at the hour t in kW

$\mathbb{P}_c = [P_{c,1},...,P_{c,t},...,P_{c;T}]$ where $P_{c,t}$ is the charging power at the hour t in kW,

$\mathbb{P}_d = [P_{d,1},...,P_{d,t},...,P_{d,T}]$ where $P_{d,t}$ is the discharging power at the hour t in kW

$\mathbf{w} = [w_1,...,w_t,...,w_T]$ where w_t is the binary variable that indicates if the battery is charging (w_t=1) or discharging (w_t=0).

$\mathbb{SOC} = [SOC_1,...,SOC_t,...,SOC_T]$ where SOC_t is the state of charge at the hour t in kWh,

SOC_0 is the initial state of charge in kWh,

W_s, W_{pv}: Annual energy sold, bought, and injected in kWh/year.

$W_{pv}^{max} = \sum_{t=1}^{T} P_{pv,t}^{max}$: Annual PV energy generated at DC busbar at standard test conditions in kWh/year.

W_{pv}^{clip}: Annual energy curtailed by clipping in the PV system (MPPT) in kWh/year.

Model constraints:

Eq. 3 and Eq. 4 correspond to the capital and operational expenditures of the PV & BESS project, respectively. Eq. 5 calculates annual earnings and expenses from energy arbitrage and contracted power charges. Eq. 6 defines the real power balance at the point of interconnection.

Eq. 7 ensures that power injected by the PV inverter is limited by the inverter capacity, efficiency, and available solar resources.

Eq. 8 describes the battery power and energy balance, accounting for distinct charging and discharging efficiencies. It also requires the final state of charge to equal the initial state of charge.

Eq. 9 specifies the size of the PV inverter in kWp. The PV inverter capacity should be enough larger to cover loads, the BESS system and the point of interconnection. Inverter Loading Ratio (ILR), also known by the DC-to-AC Ratio ($ILR = P_{pv}^{inst}/P_{pv}^{inverter}$) represents the proportion between PV installed capacity at standard test conditions and PV inverter output.

Eq. 10 specifies that injected power by the PV inverter ($P_{pv,t}$) can be lower than the available solar resource ($P_{pv,t}^{max}$). The difference between both figures ($P_{pv,t}^{max}$-$P_{pv,t}$) corresponds to the clipped power at given time t.

Eqs. 11–12 specify that charging and discharging powers must not exceed the BESS inverter capacity. Since the BESS inverter capacity and the binary variables are products of unknown variables, these constraints are *non-linear*.

Eq. 13 limits the power sold to the retailer based on feeder capacity. Eq. 14 constrains the state of charge within lower and upper limits, while Eq. 15 ensures the BESS capacity and inverter power adhere to C-rate limits. Finally, Eq. 16 summarizes annual energy traded, PV generation, and curtailment or clipping due to battery saturation or frontier capacity.

Financial indicators of the project

NPV assesses feasibility for the prosumer, who assumes responsibility for both initial CAPEX and ongoing OPEX. This analysis is supplemented by the Internal Rate of Return (IRR) and Levelized Cost of Electricity (LCOE). Note that for revenue-generating projects, a distinction must be made between standard LCOE (expenditure-only) and comprehensive metrics that include project benefits. The standard (gross) LCOE is given by:

$$LCOE_{gross} = \frac{CAPEX + \frac{OPEX_{gross}}{CRF_e}}{\frac{W_L}{CRF}} \tag{18}$$

where the OPEX only accounts operation and maintenance costs: $OPEX_{gross} = O\&M_{pv}P_{pv}^{inst} + O\&M_{bess}C_{bess}$

From the utility perspective the gross LCOE is always a positive value and it reflects the breakeven price to get a null NPV for PV investment and expenses for n years, and discount and escalation rates i and e, respectively.

2. Case Study: PV plant in Cordoba Colombia

In this section, we describe the dataset required to apply the proposed method.

The study focuses on a PV power plant located in Monteria (8.743°N, 75.882°W), Colombia (Cordoba). The facility is connected to a 115 kV AC busbar. The existing interconnection infrastructure has a maximum capacity (P_F^{max}) of 100 MW. BESS capacity (C) and inverter capacity (P_{bess}^{inv}) is 80 MWh and 20 MW.

We adopt the economic assumptions specified in the Ember benchmark [1]. The project lifespan (n) is 20 years with an annual discount rate (i) of 7.7%. The Capital Recovery Factor (CRF) is determined as:

$$CRF = \frac{i(1+i)^n}{(1+i)^n - 1} = 0.099589$$

Accounting for an annual energy price escalation factor (e) of 2.5%, the modified capital recovery factor (CRF_e) is 0.082760.

The PV investment cost (C_{pv}) is set at 436 US$/kWp, based on the following breakdown:

- Modules: 110 US$/kWp, reflecting the late 2024 price range for TOPCon modules (90–120 US$/kWp).
- Balance of System (BoS): 143 US$/kWp, covering cabling, mounting, and monitoring [2].
- Installation: 135 US$/kWp for mechanical and electrical assembly [2].

BESS investment costs are defined as C = 185 US$/kWh for energy capacity and C_{gfl} = 48 US$/kW, C_{gfm} = 48 US$/kW for the bi-directional inverter. These figures align with 2024 global estimates, though some reports suggest utility-scale costs are trending as low as 72 US$/kWh [3]. Additional costs (S_c), including BoP, BoS, design and permitting, are accounted for via a 33% multiplier ($S_c = 1.33$) applied to the total CAPEX.

The 2024 spot prices (λ) in US$/kWh are sourced from the Colombian market operator (XM)[1].

The per-unit PV generation profile ($P_{pv,t}^u$) is derived from power injection average values for 19 years PVGIS-ERAS5 dataset (2005–2023), see Figure 1. The simulation assumes optimized tilt and azimuth angles, incorporating a standard system loss factor of 19% (includes PV thermal and inverter losses). The average Hour Solar Peak (HPS) according to PV output with optimized azimuth is 4.255 hours/day.

[1] `https://www.xm.co/`

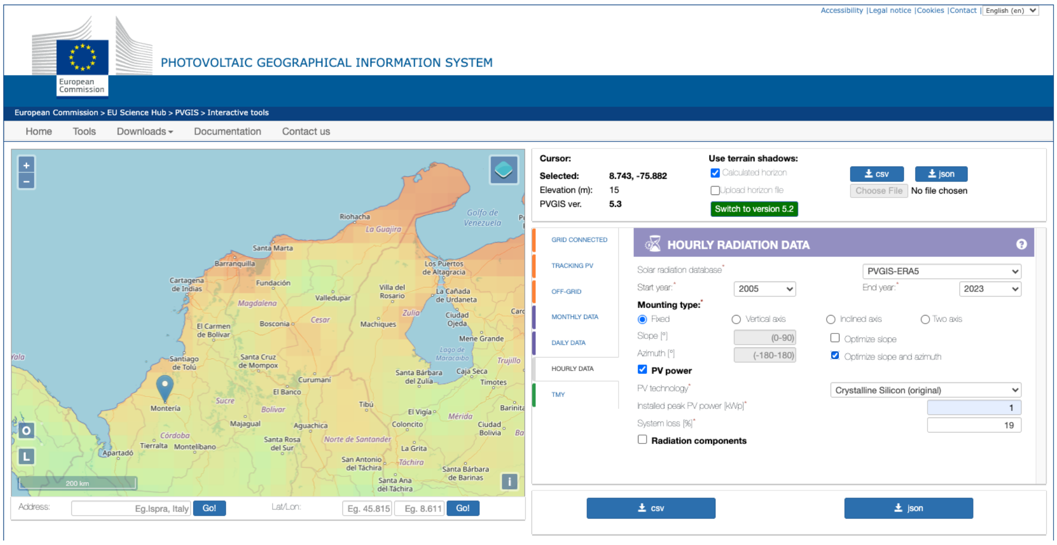

Figure 1: Portal de PVGIS `https://re.jrc.ec.europa.eu/pvg_tools/en/`

The resource pattern for Monteria, Cordoba according to PVGIS's PV output data series (2005-2023) is shown in Fig. 2

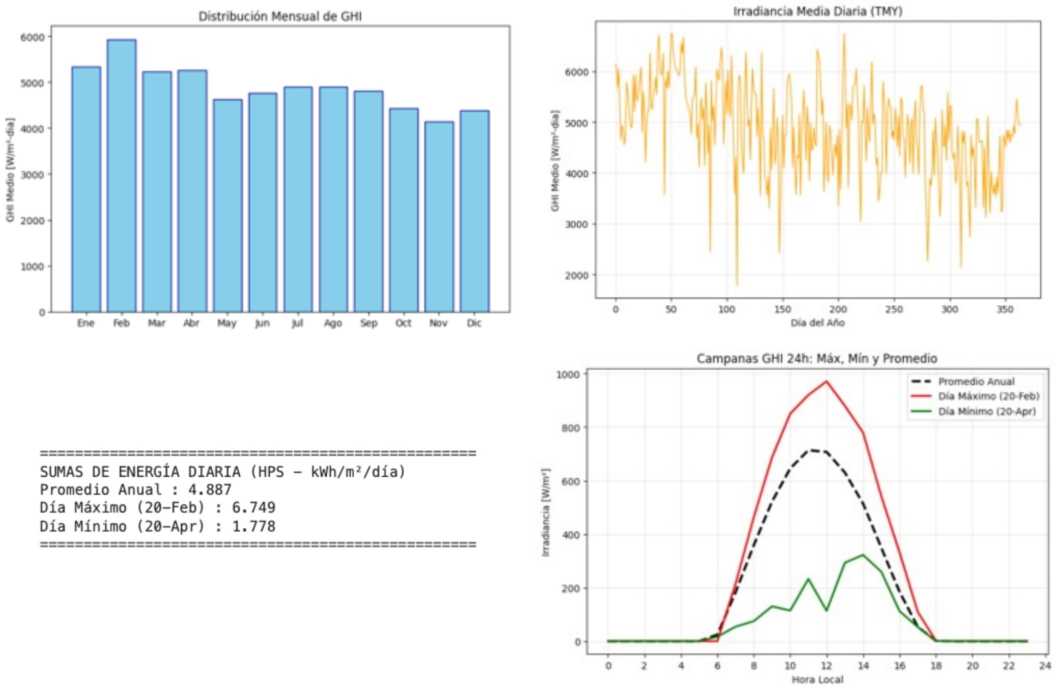

Figure 2: Portal de PVGIS `https://re.jrc.ec.europa.eu/pvg_tools/en/`

In this case the average HSP is 4.887 hours/day.

Regarding BESS operations, the Depth of Discharge (DoD) is set at 90%, constraining the State of Charge (SoC) between $\beta_{min} = 0.00$ and $\beta_{max} = 1.00$. The BESS C-rate is bounded between 0.1 and 2.0. Equipment efficiencies are defined as $\eta_c = 93.81$ %, and $\eta_d = 93.81$ %, with a round trip efficiency of 88 %. Operations and maintenance (O&M) costs are set at 12.5 US$/kWp for PV and 5.9 US$/kWp for BESS. No grid services are considered.

3. Results

The MINLP optimization model stated in Eqs. 2- 16 was scripted in Python and solved with Gurobi Optimizer version 12.0.1 build v12.0.1rc0 (mac64[x86] - Darwin 21.6.0 21H1320) in a CPU model: Intel(R) Core(TM) i7-6700HQ CPU 2.60 GHz. Scripts are available at the following GitHub page: `https://github.com/pmdeoliveiradejesus/PVutility_sizing`.

3.1. Case 1: Optimal solution with 4-h BESS of 80 MWh

Optimal solution found (tolerance 1.00e-04) in 1.24 seconds with best objective NPV = 4.453369892181e+08, gap 0.0000%. The optimization model has 24 + $9T$=78864 unknowns, 61352 continuous, 17520 integer (17520 binary), 6 + $3T$=26280 equality constraints, and 2 + $8T$=70082 inequality constraints. Model has 17520 quadratic constraints.

The optimal equipment sizing results derived from the proposed model are as follows: a PV system capacity (P_{pv}^{inst}) of 312.7 MWdc. This PV capacity corresponds to 622,203 modules (500 W each). PV inverter load ratio (ILR, MWdc/MW): 2.61.

The annual energy balance is illustrated in Figure 3. The model ensures that the condition $W_s = W_d - W_c + W_{pv}$ is strictly met.

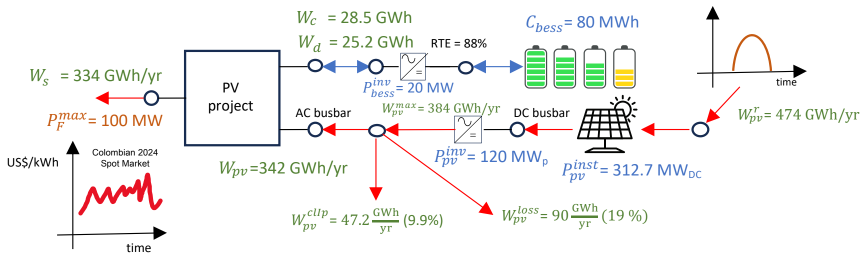

Figure 3: Optimized annual energy balance showcasing generation, storage, and grid exporting ratios.

Fig. 3 illustrates the annual energy balance.

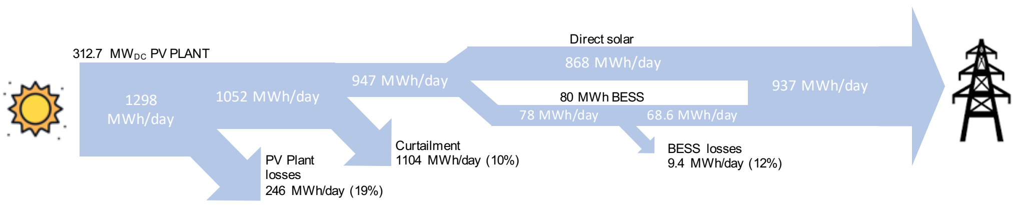

Figure 4: Sankey diagram of the project (average day).

Fig. 4 shows the average daily Sankey diagram of the system.

Figure 5 provides a granular view of the BESS behavior during typical weeks in January and July.

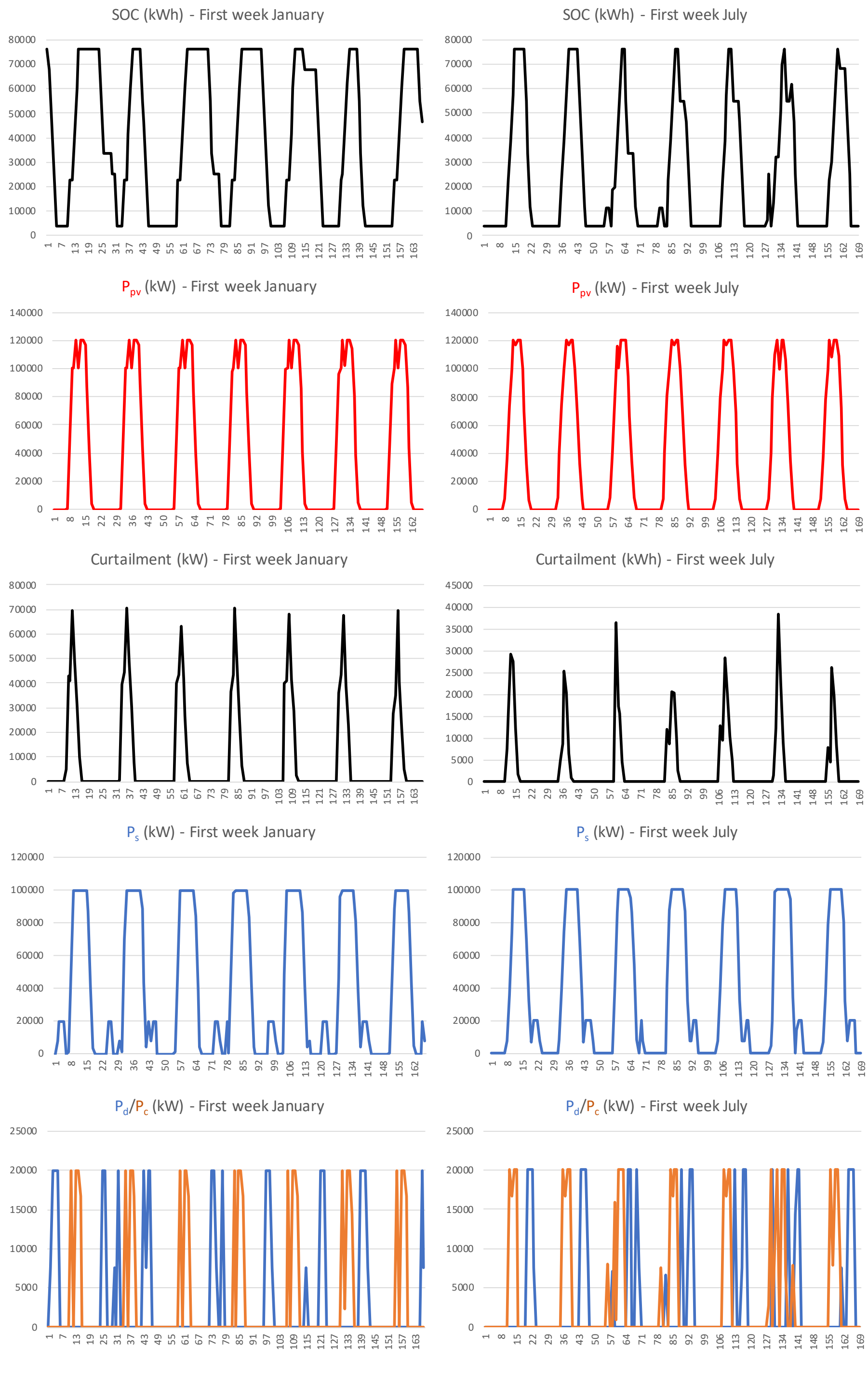

Figure 5: BESS State of Charge (SoC) and power dispatch behavior during 1st week January and July

It can be observed that the generation curtailment is significant, at 10 % of total generation, due to the limitation imposed by the boundary (100 MW) when the battery is charged. To maximize the use of the boundary and thus maximize sales, it is necessary to incur this loss, which, viewed as an opportunity cost, could be used for other purposes, such as green hydrogen.

Figure 6 provides a granular view of the BESS behavior during first three in January.

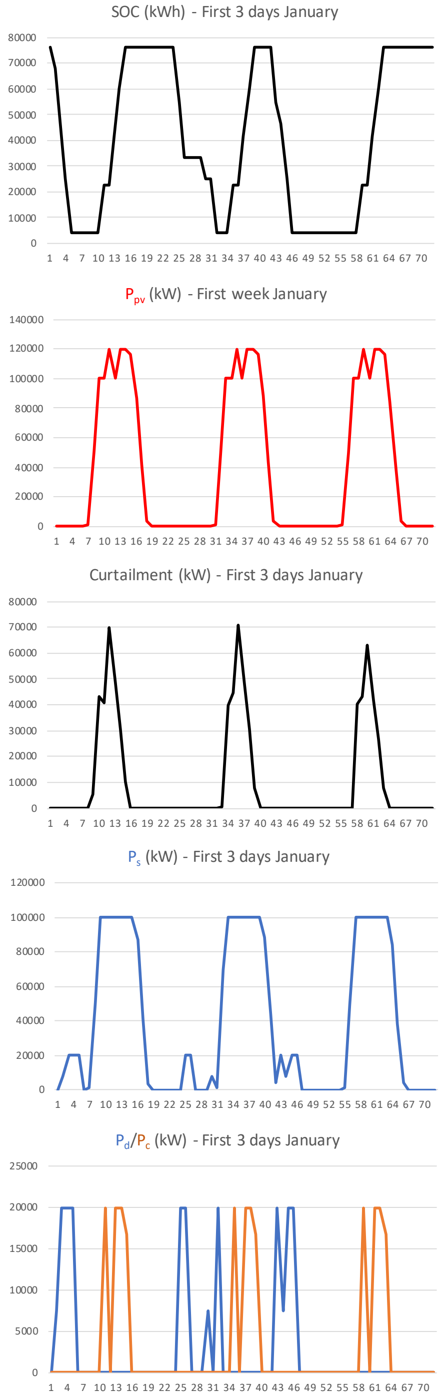

Figure 6: BESS State of Charge (SoC) and power dispatch behavior during 1st three days in January

Another interesting aspect is that the battery doesn't discharge within 20 to 24 hours. The optimizer chooses the best time to discharge, depending on the spot price, known as the day-ahead price.

Operational Expenditure with the project (OPEX): 4.3 million US$/year

Energy earnings of the project (Es): 56.5 million US$/year
Capital Expenditure (CAPEX): 188.5 million US$
BESS battery cost: 14,800,000.00 US$
BESS inverter cost: 960,000.00 US$
Additional Costs: 60.3 million US$
PV system cost: 127 million US$
Net Present Value: 445.3 million US$
Project Cash Flow: 52.4 million US$/year
Internal Rate of Return: 27.6 %
Pay Back Time: 4.4 years
LCOE: (only CAPEX and OPEX) 75 US$/MWh
Benefit-Cost Ratio 2.79
Sensitivity Analysis:

Figure 7 shows how tha NPV, IRR, PBT and the LCOE changes when the PV capacity changes from 100 to 1000 MW.

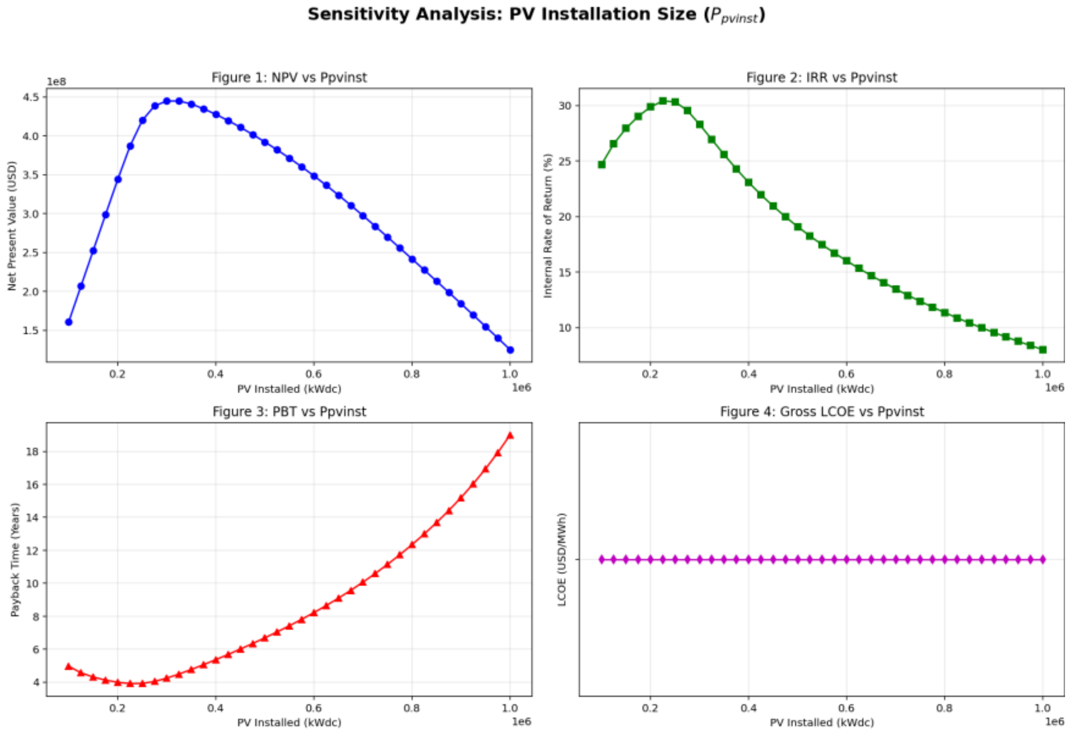

Figure 7: Sensitivity analysis varying P_{pv}^{inst} from 100 to 1000 MW

Notice that the best (NPV 445.3 million US$) is obtained with P_{pv}^{inst} = 312MW.

3.2. *Case 2: Optimal solution, all PV and BESS equipment*

Optimal solution found (tolerance 1.00e-04) in 11.84 seconds with best objective NPV = 6.908203840348e+08, gap 0.0000%. The optimization model has 24 + 9T=78867 unknowns, 61355 continuous, 17520 integer (17520 binary), 6 + 3T=26280 equality constraints, and 2 + 8T=70082 inequality constraints. Model has 17520 quadratic constraints.

The optimal equipment sizing results derived from the proposed model are as follows:

PV system capacity (P_{pv}^{inst}) of 732 MWdc. This PV capacity corresponds to 1,456,744 modules (500 W each).

BESS capacity C_{bess}: 1,456,460.33 kWh

BESS inverter capacity $P_{bess}^{inverter}$: 242,911.24 kW

PV System inverter $P_{pv}^{inverter}$: 342,911.24 kWp

BESS C-rate: 0.17 1/h, 6.00 hours, PV inverter load ratio (ILR, MWdc/MW): 2.13

The annual energy balance is illustrated in Figure 8. The model ensures that the condition $W_s = W_d - W_c + W_{pv}$ is strictly met.

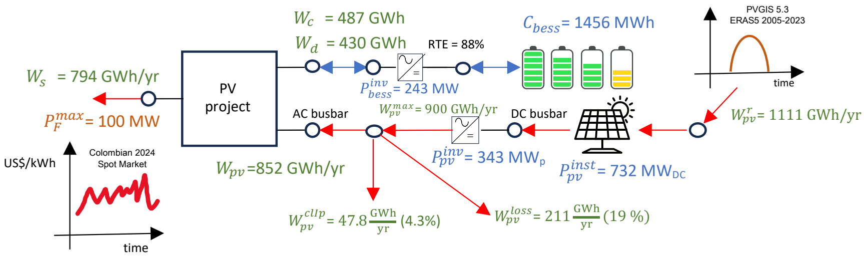

Figure 8: Optimized annual energy balance showcasing generation, storage, and grid exporting ratios.

Fig. 8 illustrates the annual energy balance.

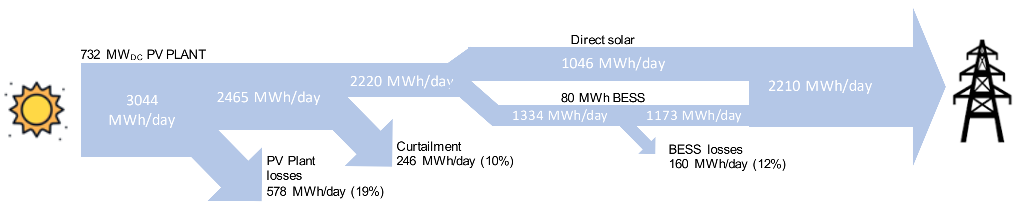

Figure 9: Sankey diagram of the project (average day).

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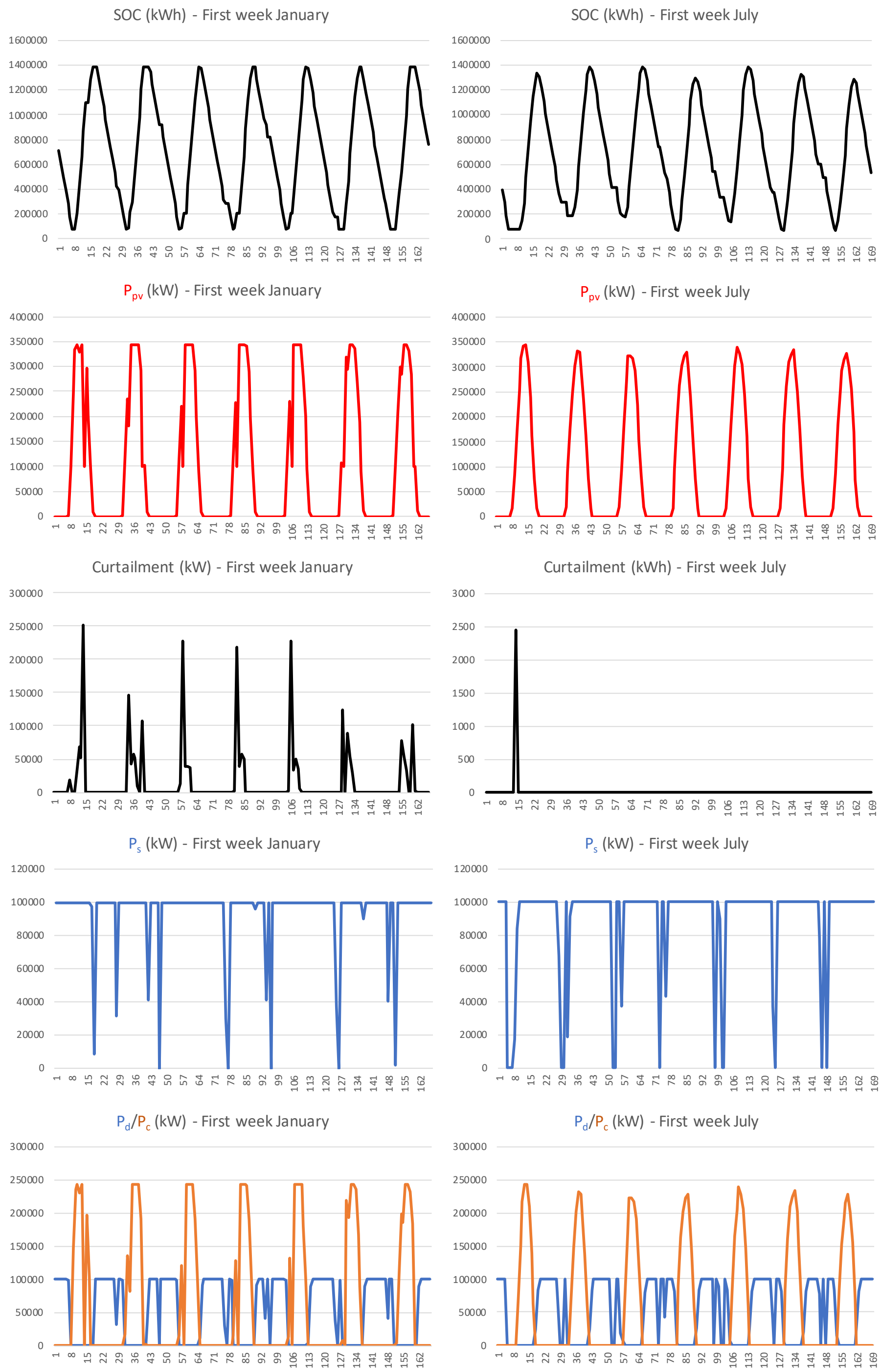

Figure 10: BESS State of Charge (SoC) and power dispatch behavior during 1st week January and July

It can be observed that the generation curtailment is significant, at 4.3 % of total generation, due to the limitation imposed by the boundary (100 MW) when the battery is charged. To maximize the use of the boundary and thus maximize sales, it is necessary to incur this loss, which, viewed as an opportunity cost, could be used for other purposes, such as green hydrogen.

Figure 11 provides a granular view of the BESS behavior during first three in January.

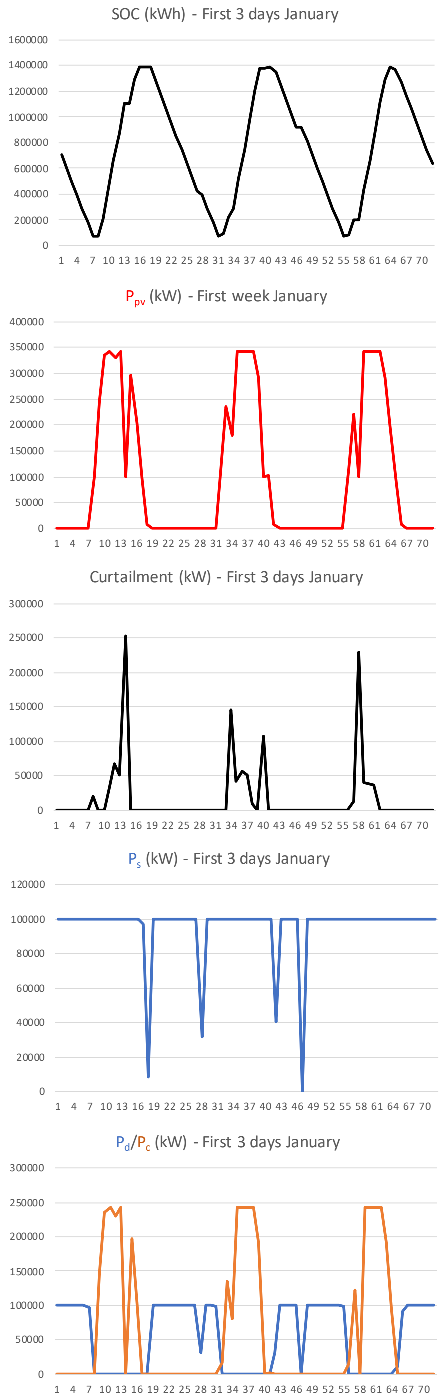

Figure 11: BESS State of Charge (SoC) and power dispatch behavior during 1st three days in January

Another interesting aspect is that the battery doesn't discharge within 20 to 24 hours. The optimizer chooses the best time to discharge, depending on the spot price, known as the day-ahead price.

Operational Expenditure with the project (OPEX): 17.7 million US$/year

Energy earnings of the project (Es): 138.4 million US$/year
Capital Expenditure (CAPEX): 767 million US$
BESS battery cost: 269 million US$
BESS inverter cost: 11.6 million US$
Additional Costs: 245 million US$
PV system cost: 300 million US$
Net Present Value: 690 million US$
Project Cash Flow: 120 million US$/year
Internal Rate of Return: 14.7 %
Pay Back Time: 9.0 years
LCOE: (only CAPEX and OPEX) 123 US$/MWh
Benefit-Cost Ratio 1.58
Sensitivity Analysis:

Figure 12 shows how tha NPV, IRR, PBT and the LCOE changes when the PV capacity changes from 100 to 1000 MW.

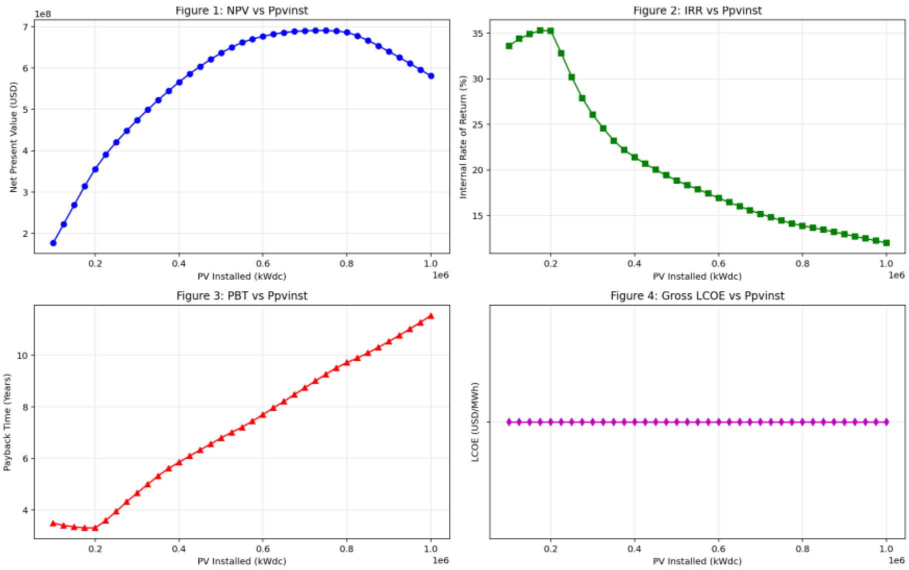

Figure 12: Sensitivity analysis varying P_{pv}^{inst} from 100 to 1000 MW

Notice that despite the best NPV is 732 million USD$, the NPV curve is almost flat since P_{pv}^{inst} greater than 600MW. A good solution is installing 600 MW (sub-optimal solution) yielding better IRR and PBT.

4. Conclusion

This document shows the results of a sizing model for a utility-scale PV plant located in Monteria, Cordoba, Colombia. The method determines the optimal size of the PV plant in MWdc (number of PV modules) and MW (size of the PV inverter) considering a fixed. The model also determines the size of the Battery Energy Storage System. The project is funded in the context of wholesale spot electricity markets where PV & BESS system's energy curtailed can be used to arbitrage. The method aims to get the PV plant dispatch (including curtailment) and the BESS scheduling such that maximize the Net Present Value of the project using Ember's economic figures [1]. The method takes advantage of BESS trading Colombian spot prices (2024) provided by XM in order to leverage the project. Results are obtained for two maximum NPV conditions in Monteria considering a point of interconnection of 100 MW: 1) determining the PV's MWdc by fixing a 80 MWh BESS system with C-rate of 0.25 (4 hours), and 2) determining PV's MWdc and BESS specification. For the first case, results show that to maximize the project NPV (445 million

US$) it is necessary to install a 312 MW PV power plant with good financial KPI's: IRR of 27.6 %, a payback of 4.4 years, and a LCOE of 75.0 US$/MWh. For the second case, results show that to maximize the project NPV (690 million US$) it is necessary to install a 732 MW PV power plant with the following KPI's: IRR of 14.7 %, a payback of 9 years, and a LCOE of 123 US$/MWh.

References

References

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